

1) What is Web 3.0, and how is it different from Web 1.0 and Web 2.0?

Web3 (or Web 3.0) is the third generation of the World Wide Web, designed to be a decentralized, trustless, and user-owned internet. While Web1 was "read-only" and Web2 is "read-write," Web3 introduces the concept of "read-write-own," where users retain control over their data, digital identity, and assets using blockchain technology

2) What are the functional differences between the front end of a web application and its back end?

The frontend and backend of a web application serve as the two halves of a single system, often compared to a restaurant's dining room (frontend) and kitchen (backend).

The frontend is everything a user sees and interacts with directly in their browser and the backend consists of the "hidden" machinery that powers the frontend from a server.

3) In your own words, explain the process that takes place from when you type a URL into the address bar in your browser until you finally view the page you have requested. Include the HTTP request-response cycle in your answer and provide an example of the HTTP request and response messages.

The Journey of a URL:

1. The process begins when you provide a client (like a browser) with a URL. This URL acts as a human-friendly address for a resource sitting on a server somewhere in the world.
2. The browser uses that URL to identify the protocol (usually HTTP), the domain name (the server's address), and the specific path to the file you want. It then builds an HTTP Request message.
3. The browser sends this request to the web server.
4. The server receives the request and uses the information inside it to build a response. If the page is dynamic (Web 2.0), a program behind the scenes might extract personal data from a database to customize the page for your user account.
5. The server sends an HTTP Response message back to your browser.

6. Finally, the browser interprets the HTML code in the response and renders the page on your screen for you to view

Example HTTP Request

An HTTP request starts with a request line followed by header lines:

```
GET /bootcamp/web_dev.html HTTP/1.1
Host: www.hyperiondev.com
User-Agent: Mozilla/5.0
Accept-language: en
```

Example HTTP Response

An HTTP response starts with a status line, followed by header lines, and finally the entity body (the content)

```
HTTP/1.1 200 OK
Date: Tue, 14 Nov 2023 15:35:05 GMT
Server: Apache/2.2.3
Content-Type: text/html
Content-Length: 6821

<!DOCTYPE html>
<html>
  <body><h1>Hello, World!</h1></body>
</html>
```